

Investment Subsidy NWO Medium Application 2008

Please indicate by which NWO Division your application should be reviewed (several options possible).

ALW ? CW ? EW ? GW ● MaGW ? ZonMw ?

1. Applicants(s)

a. *principal applicant / contact*

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b. *Researchers involved in the application*

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Prof. dr. Antal van den Bosch	Machine learning, WSD	Tilburg centre for Creative Computing, Faculteit Geesteswetenschappen, Universiteit van Tilburg
Prof. dr. Maarten de Rijke	Machine learning, WSD	ISLA, Faculteit der Natuurwetenschappen, Wiskunde en Informatica, Universiteit van Amsterdam

c. Does the local authority support your application? ● Yes (did you inform your superior and accepts your institute/university the conditions for support by NWO?) YES

Relevant authority
 Name: G.J.M. Nijsten
 Position: Director of the Faculty of Arts

2. Project title

DutchSemCor: a Dutch corpus tagged with the word senses from the Cornetto database

3. Project summary (max. half page)

Much work in modern linguistics is facilitated and driven by annotated text corpora: collections of texts with manually labeled structures and interpretations. In recent years, the annotations used in these corpora have become increasingly rich and semantically informed. Yet, there is still no large semantically annotated text corpus for Dutch. The goal of DutchSemCor is to deliver a one-million word Dutch corpus that is fully sense-tagged with senses and domain tags from the Cornetto database (STEVIN project STE05039). 250K words of this corpus will be manually tagged. The remainder will be automatically tagged using three different word-sense-disambiguation systems (WSD), and will be validated by human annotators. The corpus data will be based on existing corpus material collected in the projects CGN, D-CoI and SoNaR. These corpora have already been automatically annotated with morpho-syntactic tags and structures. The corpora will be extended where necessary to find sufficient examples for meanings of words that are less frequent and do not appear in the above corpora. The resulting corpus, for which we aim to offer the same balance in types of text as these basic resources, will be extremely rich in terms of lexical semantic information. Its availability will enable many new lines of research and technology developments for the Dutch language. In particular, it

will enable research into the relation between language form and language interpretation, and as such it will be applicable in the fields of cognitive science, (psycho-)linguistics, language learning and language teaching, semantic web applications, information retrieval, machine translation, text mining, and document interpretation (summarization, topic segmentation). We foresee that the corpus will create new directions of research and technology development on a par with current developments for English. Moreover, given the lessons learned from recent annotation efforts on English and other language corpora, developing the corpus will lend us the opportunity to genuinely advance the state of the art in corpus sense-tagging.

4. Keywords (maximum of 5)

Text corpora, Semantic annotation, Lexical semantics, Linguistic resource

5. Is this a revised application?

No, but preceded by the NWO Middelgroot *startsubsidie* project DutchSemCor (NWO project 380-70-010) that was carried out as a preparation to this proposal.

6. Has this application been submitted elsewhere?

no

The **maximum** length of items 7 to 11 is **10** pages with minimum font 10pts.

7. Detailed description of the research plan and investment

7.1 Goal

The goal of this project is to deliver a one-million word corpus with sense tags that represent one hundred (100) examples of all the different meanings of 3,000 of the most frequent and most polysemous nouns, verbs, and adjectives in Dutch. The sense tags are taken from the senses in the Cornetto lexical database, recently built within the STEVIN framework. These 3,000 words have about 3.4 meanings on average per word, and represent the most difficult and complex words of the language, accounting for more than 80% of all the word occurrences in texts. They have earlier been analyzed manually and with great care in the Cornetto project (project STE05039). The Cornetto project resulted in a rich semantic lexical database for Dutch.

Two hundred and fifty thousand word occurrences of this corpus (250,000 tokens) will be manually tagged; the remainder (750,000 tokens) will be automatically tagged using three different *word sense disambiguation* (WSD) systems, of which the output will be validated by human annotators. The resulting corpus will be used by the linguistic, cognitive, and language engineering research communities.

The corpus will be balanced, in the sense that it will contain a wide selection of different registers and types of text and transcribed speech, inheriting this selection from the corpora on which it will be based (the Spoken Dutch Corpus (CGN), the D-CoI pilot corpus, and the SoNaR corpus as it will become available). The corpus will also be balanced semantically, i.e., all the different meanings of the above words will get at least one hundred examples. This will necessitate the mining of other sources such as the Internet to find sufficient examples for rare meanings. The specific rationale for the selection of the words to be tagged, the amount of corpus data to be covered, and the specific method of tagging is given in the Appendix of this proposal, which includes a report on the pilot tagging exercise and the workshop on semantic tagging carried out in the NWO Middelgroot *startsubsidie* for DutchSemCor (NWO project 380-70-010).

Our manual tagging efforts will be further guided by the following considerations:

1. We will focus on the 3,000 most frequent, most polysemous and domain independent nouns, verbs and adjectives. The ratio of nouns, verbs and adjectives will be 9:2:1;
2. We will exclude tagging of multiword phrases such as separated occurrences of phrasal verbs (“uitbreken”, “aanbreken”) and fixed collocational constructions (“een lans breken voor”);
3. We will exclude metaphoric and metonymic uses that are not in the Cornetto database;
4. We will include some domain specializations as illustration.

The motivation for the focus is that the most-frequent, most-polysemous and domain independent words are the most difficult to handle, yet they occur in almost any type of text. Even in the most domain-specific types of text, words such as “breken,” “zaak” and “plaats” (*break*, *case*, and *place*) tend to (partially) keep their polysemy as none of their senses belongs exclusively to a domain. This choice is directly linked to the intended general applicability of the automatic WSD methods to be developed alongside the corpus.

The motivation for excluding the tagging of collocational constructions is that this would require resources that do not exist as yet: (1) a means to automatically detect collocations (as investigated for Dutch in the Stevin project IRME) and (2) representations of meanings of collocations in the Cornetto database, which does yet not represent these meanings separately. The same holds for metaphoric and metonymic uses of words that are not yet listed in the database. The availability of a DutchSemCor corpus will however be an ideal stepping stone and test bed for the further development of automatic collocation detection methods; this is one of the research areas in which we envision that DutchSemCor will enable new research. This being said, a search for a word such as “breken” (*break*) will nevertheless result in many occurrences related to multiword expressions that include “breken” and not cases of the isolated word. Automatic detection of possible multiword uses could be used to exclude these multiword uses from isolated cases. Figurative use (“haar ogen braken”, *her eyes broke*) and metonymic extensions (“die twee tosti’s moeten nog betalen”, *the two ham-cheese sandwiches still need to pay*) cannot be filtered automatically. They either need to be skipped or tagged with a figurative or metonymic usage flag. Other ‘new’ meanings that are not derivative or related to creative language use, on the other hand, need to be added to the database and subsequently tagged with the new

sense. Special strategies will be agreed for undecidable cases (e.g. where the context leaves open the sense ambiguity).

Although we focus on domain-independent “hard” WSD, we will devote some attention and effort to domain-specific WSD, as the recent literature on WSD has to a large extent focused on it. The positive conclusions from most of this work is that domain-specific WSD is feasible (Magnini et al. 2002, Koeling et al. 2005, Buitelaar et al. 2006), arguably because it avoids solving the hard domain-independent cases.¹ We will include some domain-specific sense annotation in the corpus to illustrate the application of the above-mentioned related work to Dutch, but overall we expect that this issue can partly be solved without a DutchSemCor corpus, e.g., by automatically acquiring (unsupervised) domain corpora. These solutions are less language-dependent as well. For the domain-tagging, we will define two specialized subcorpora. These corpora will be fully sense-tagged, i.e., all words in the text will be tagged as they occur in the text. This tagging will also include named-entity tagging and normalization. Named entity-tagging can be done mostly automatically, but since named-entities hardly occur in Cornetto, we will extract their types from Wikipedia and match these with Cornetto synsets; in addition, we will map synsets in Cornetto to Wikipedia articles and categories thus creating a re-usable Cornetto-to-Wikipedia mapping.

In addition to the tagged corpus itself, we will also develop a variety of sense-taggers or word-sense-disambiguation (WSD) systems based on different techniques: *knowledge-rich*, *supervised data-driven*, and *unsupervised data-driven*. These sense-taggers are used within the project:

1. The *knowledge-rich tagger* (WSD-1) will make use of existing resources, specifically the Cornetto database. This database, the product of many years of lexicographic expert work, offers selected glosses and example sentences as well as rich semantic and domain constraints, which could be used as prototypical contexts in a baseline sense disambiguation system.
2. As soon as a sufficiently large amount of corpus data has been manually annotated, *supervised machine-learning-based* WSD modules can be trained (WSD-2). In the case of WSD, “sufficiently large amounts” are typically measured per polysemous word, for the development of so-called word expert modules; in certain cases, as few as a handful of examples per sense already provide better performance than guessing the most frequent sense.
3. The past decade has produced several interesting *unsupervised* methods. For instance, one type of approach uses a bilingual corpus (e.g., Open Source corpora such as EuroParl) to annotate senses that in the other language are realized as different words. Another type of approach uses word vector clustering to discover that a single polysemous word occurs in more than one semantically coherent cluster. These techniques will be used in WSD-3.

Finally, we will develop a dedicated sense-annotation tool that interacts with the Cornetto database. This is a tool with which a human annotator can select senses for words occurring in text. All the results will be freely available as open source under GPL, only limited as far as the input corpora are limited. No further restrictions are imposed.

7.2 Methodology

As a general methodology of the project, we use a mixture of automatic and manual tagging procedures. The motivation for this approach is based on the outcomes of the DutchSemCor workshop held in May 2008, and the pilot tagging exercise, both carried out as part of the study funded by the NWO Middelgroot *startsubsidie* awarded to DutchSemCor (details can be found in the Appendix).

The envisaged corpus of 1 million tokens will be split into two parts that are handled in different ways. The first part of about 250,000 tokens is to be tagged in a traditional way: on average 25 examples of each meaning of the 3,000 words are searched and tagged by two human annotators. This tagging is supported by a knowledge-rich tagging system that does not rely on training examples. The second part of the corpus will cover 750,000 tokens, adding another 75 examples for each word meaning. The coverage of the corpus is partly based on the remainders of the general corpora used, and partly on the necessity to find sufficient examples for each meaning of the selected words. It will also include the two domain specializations. This corpus will be tagged fully automatically using tagging systems that are trained by the manually tagged data

¹ Magnini et al. report 80% precision for the domain-based disambiguation in Senseval-2 but their system has a recall of only 20%, since it is restricted to domain-specific word meanings only.

acquired so far and any other data that can be used. The manual tagging in the second phase then involves validating the automatic assignments by a single annotator. This means that we focus on those cases where the confidence of the system is low and different systems disagree, as in active learning or co-training methods. Not that we can also group word occurrences based on their estimated meanings and compare the different contexts in which they occur. If there are insufficient examples of a word in a particular meaning in the corpus, sampling techniques can be used to find additional examples of the word in its context, e.g. on the web or in large textual corpora (cf. the extraction of Topic Signatures, Agirre & Lopez de Lacalle 2004).

At the end of this section we include a table with the effort allocation and time schedule for the different work packages (see also the Appendix for a more detailed motivation). Next, we will discuss each work package in more detail.

WP1. Corpus selection and preparation

We will first make a proper selection of text subcorpora, to be extracted from existing corpora. Candidates are: the Spoken Dutch Corpus (CGN), D-CoI, SoNaR, Twente News Corpus, Wikipedia, and the Schrooten & Vermeer children's book corpus (Schrooten & Vermeer, 1994). Most of these corpora have been morpho-syntactically tagged (Twente News Corpus and Wikipedia are automatically tagged; Schrooten & Vermeer uses a blended semantic-syntactic tagging system) and are formatted according to the D-CoI guidelines.

Table 1: Potential corpora for DutchSemCor

Corpus	Size	Annotation levels	Availability
CGN	9 million words	Manual PoS tags and lemmas, syntactic parses for 1 million words	Licensed by TST Centrale ²
D-CoI	65 million words	Automatic PoS tags, lemmas, and syntax	Licensed by TST Centrale
SoNaR	500 million words (projected)	Automatic PoS tags, lemmas, and syntax	Licensed by TST Centrale (future)
Twente News Corpus	450 million words	Automatic PoS tags, lemmas, and syntax	Limited research license
(Dutch) Wikipedia	50 million words (approx.)	Automatic PoS tags and lemmas	Free
Schooten & Vermeer	1.8 million words	Manual syntacto-semantic tags	Limited research license

In addition to the tagged corpora, we use large untagged corpora by extracting words in a limited “snippet” context used in a specific meaning from the Web; so-called topic signatures. The results of these corpora can be used to apply WSD to the tagged corpora or to derive feature bundles. Copyright issues are avoided by restricting snippets to at most 155 characters that could be re-distributed, as declared in the legal document:

<http://www.rechtspraak.nl/Gerechten/Rechtbanken/Alkmaar/Actualiteiten/Twee+uitspraken+in+kort+ge+ding+betreffende+Jaap.nl.htm>

The snippets are especially needed to acquire sufficient examples for word meanings that are less frequent.

The selection of the text from the corpora is further determined by the following criteria:

1. Availability for distribution of the text and the semantic tags without licensing constraints;
2. Availability for distribution of the text through the TST-Centrale (licensed) and of the semantic tags with offsets (for free) by the consortium and the TST-Centrale;
3. Integration of the semantic tags in existing morpho-syntactic annotations, preferably validated;
4. Balanced in terms of genres, modality, domains and topics;
5. Balanced in terms of distribution of word meanings;
6. Formatted and structured according to the most recent Corpus Annotation Guidelines and the practice used in current text corpus projects;

² TST Centrale: Dutch HLT Agency, <http://www.tst.inl.nl>

Specifications for the XML format of the corpora needs to be extended so that the sense-tags from the Cornetto database can be included. For this, we follow de facto standards for sense annotation and the Semantic Annotation Framework (ISO/CD 24617-1, TC 37 - Terminology and other language and content resources, <http://www.iso.org>). The structure will enable storage of multiple weighted tags for multiple senses and proposed by different systems, including tags manually assigned by human annotators. We will extend the DTDs for the current corpora to accommodate the tagging information. As a pre-processing step, we will assign a domain classification to the text. These domain tags can be used to group occurrences of words per dominant domain, which are also more likely to share the same meaning of the word and to acquire selections of the corpus for word meanings that are dominant only in specific corpora. WordNet domain tags (Magnini et al. 2002) are already added to the meanings of each Dutch word in the Cornetto project. Importantly, sense-tagging is intrinsically different from morpho-syntactic corpus annotation. The latter use a limited set of tags, and the tagging structures are applied to complete spans of words in the sequence in which they occur in utterances or sentences. In the case of semantic tagging, understanding the different meanings of a word (as an unlimited tag set) takes an annotator a substantial amount of time, especially since this process needs to be repeated per polysemous word. The most efficient way for sense tagging is therefore so-called targeted tagging (Fellbaum et al. 2001), i.e., to handle all occurrences of a single word at the same time. For targeting tagging, we need to build an index on the lemmas in the text, possibly in combination with contextual and domain features that apply to each occurrence. The human annotators can then retrieve all occurrences of a word with a query and assess its different meanings with a limited cognitive effort. The word occurrences can be grouped on shared left and right contexts (word, part-of-speech or dependency-based), on the basis of domain tags, or on already assigned or suggested sense-tags. This was shown to be very effective in the pilot study. Special indices and retrieval services will be developed to allow such access. We will benefit here from the rich annotations and structural analysis that is available in the currently available corpora for Dutch.

WP2. Sense Annotation Tool

Given the indices on lemmas in context and the retrieval services, a sense-annotation tool needs to be developed that gives access to these data and the Cornetto database. Existing sense-annotation tools depend on the type of indices and the resource that is being used. This is different for each language group. For example, the Basque and Spanish wordnet groups used the WEI database (Agirre et al. 2006) for accessing their Basque and Spanish wordnet and other resources. Furthermore, they built the Basque wordnet at the same time as they tagged the corpus. This means that the wordnet changed continuously during the tagging, which requires intense collaboration across the team of annotators. In our case, a large wordnet database is already available which is also enriched with other semantic layers (Wordnet Domains and the SUMO ontology). This makes it easier to work independently and it makes it possible to support and validate manual annotations by automatic tools. To be able to deploy an existing sense-tagging system it needs to be adapted so as to assign sense-tags from the Cornetto database and to interface with this database. For the purpose of targeted tagging, all occurrences of a word need to be displayed in a KWIC index (targeted tagging) and interfaced with the meaning specification in the Cornetto database. Special measures will be taken to detect and exclude multiword usages of words from the retrieved text. Furthermore, if a word in the corpus does not occur in Cornetto, it will be labeled by the human annotator as a new word meaning and added to the database during an editorial round. Similarly, the sense-annotation tool supports labeling figurative usage, metonymic usage and multiword phrases. It should also be able to rank senses on different strategies: randomly, by sense, by tagging confidence or (dis)agreement of automatic WSD systems, etc. Hence, the sense-annotation tool needs to be able to take the lemmatized form and part-of-speech into account, and load automatically assigned senses with confidence scores. Furthermore, the sense-annotation tool needs to take the lemmatized form and part-of-speech into account, and load automatically assigned senses with probabilities for confirmation. The annotation system should have direct access to the Cornetto database, to visualize the available senses and allow editing of the database by the human annotator with the current Cornetto editing tool. There should also be a possibility to tag running text, i.e., tag all occurrences of a word in a domain-specific sub-corpus. A pilot annotation tool (DutchSemCor tagger 1.0) was developed by the VUA for the pilot carried out in the DutchSemCor preparatory project. This system will be used as a basis for developing the annotation tool in this project. The pilot tool is a stand-alone application that uses search indexes from Irion Technologies to access samples from the corpora that were available at that time. The system does not store the tags in the corpus but just logs the actions of the annotator in a simple log file. The newly developed system needs to be web-based and to use fast direct access to corpus data, incorporate rankings and votings from automatic sense tagging systems, extensively log the actions of the annotators and store the tagging directly in the corpus structure. The web-based client

should be usable by a wider community of people in the NLP field, which implies keeping track of the status of the human annotators and possibly derive inter-annotator reliability ranks. Finally, for post-processing and editorial discussions, we need reporting and monitoring tools to provide feedback to the annotators.

WP3. Automatic tagging of subcorpus-1

A baseline classifier is trained with the rich information in the Cornetto database and any other semantic databases that are linked to the database (Atserias et al. 2004). This classifier will be based on an existing unsupervised classifier that can use a knowledge base such as Cornetto. This classifier is used as a WSD system to make a first selection of probable senses in the subcorpus-1. This system is called WSD-1.

WP4. Manual verification and correction of the subcorpus-1

In the first phase, we will review the annotation guidelines of the current corpora for Dutch and international de facto standards for semantic tagging. We will develop the specific guidelines and protocols for DutchSemcor and make these available through the web forums for human tagging. Special attention will be paid to exclude more difficult phenomena such as multiword phrases, metaphors, and metonymy. These will not be resolved but need to be labeled in such a way that they are excluded from the tagging or are detectable. We will develop simple solutions to handle these cases without hampering the more basic tagging that we want to focus on. It should enable future-research, but it should not lead to undirected tagging in the current project. All these cases will be exemplified and we will provide users access to similar cases handled during the project. Using the corpus annotation tool developed in WP2, linguists/lexicographers will manually assign senses of subcorpus-1 (250K words), where possible supported by WSD-1 if it yields confident results. If there are no suggestions, the annotator will complement the system. We expect that the tagging will also lead to improvements to the Cornetto database, where senses may be missing or sense-distinctions may need to be adjusted. This will be done through the current editing system that is already available for Cornetto. The tagging is carried out twice by two different persons to acquire information about inter-annotator agreement. Volunteer tagging will be organized to capture some of the power of the web-community. All tagging activities will be discussed by the editorial team in bi-weekly sessions lead by the post-doc, and the results of these discussions will be incorporated in the guidelines and the tagging forum.

WP5. Automatic tagging of subcorpus-2

Subcorpus-1 can be used to train existing supervised WSD systems of UvA (WSD-2) and UvT (WSD-3). Subcorpus-2 (500k words) is then tagged using WSD-1, WSD-2, and WSD-3. The results of each system will be stored together, so that they can be accessed separately and in combination. WSD-2 and WSD-3 represent different algorithmic approaches (supervised vs. unsupervised) as well as complementary research strands in semantically informed language technology. Based on entity models, i.e., language models built up from local contexts (typically web snippets mentioned earlier), the WSD system developed at UvA is heavily focused on named entities and builds on named entity taggers trained on CoNLL data and purpose-built training material (developed in house at UvA). The WSD system developed at UvT is based on a word expert architecture. Each word expert is a memory-based classifier that has been co-optimized on feature selection (drawing on a pool of both local neighborhood and global keyword features), and algorithmic parameter optimization. This system, which won the “English all-words WSD task” at Senseval-3, uses extensive amounts of computational power during training and optimization, but this effort can be parallelized on any multi-processor or grid computing environment. Combining systems trained on different types of evidence gives better results than any separate system.

WP6. Manual verification and correction of the subcorpus-2

The linguists-lexicographers will manually correct and complement subcorpus-2 and where necessary adjust the Cornetto database. No double tagging will be applied to this corpus. The tagging will focus on validating the automatic systems in case of low confidence scores or disagreement across systems. Furthermore, the manual tagging is directed to find examples for low-frequent senses, the domain extensions and to do some full text tagging.

WP7. Automatic tagging of subcorpus-3

Subcorpus-1 and 2 can be used to train existing supervised WSD systems of UvA (WSD-2) and UvT (WSD-3). Subcorpus-3 (250k words) is then tagged using WSD-1, WSD-2, and WSD-3. The results of each system will be stored together, so that the separate results can be accessed and the different combination can be used. Combined votes will again direct the human effort in the last round.

WP8. Manual verification and correction of part of the subcorpus-3

The linguists-lexicographers will again manually correct the automatically tagged words of subcorpus-3 (250k words). Again no double tagging is required and the focus will be on poorly scored cases and lesser represented word meanings.

WP9. Documentation and delivery

The corpus and the WSD systems will be delivered to the TST-Centrale, with proper documentation. The documentation will include inter-annotator agreement and confidence scores for the automatically disambiguated part. The WSD systems and the annotation tool will be available through GNU-GPL licenses from the partners in the project. The WSD systems, the annotation tools and the tagged corpus will be available from the TST-Centrale (Dutch HLT Agency). Subcorpora that have a particular license governed by the TST Centrale will remain licensed under same conditions. No new negotiations for licensing will be carried out in this project. If possible, we will distribute the semantic tags and the offsets of the text for free through the TST-Centrale and the consortium, whereas the text itself can be obtained from the TST-Centrale through license. Obviously, if the text can be distributed freely, we will include it as well (see above for the corpus selection and preparation details in WP1). Statistics on sense-distribution and generalized contextual data will be stored in the Cornetto database and distributed through the TST-Centrale in a new release. All direct results of the project will be available free of charge.

During the project, early distributions of data are available to the public. We also foresee voluntary contributions from the research community outside the project. Such contributions will be registered in such a way that they can be discriminated and validated separately from the official project effort.

Deliverable	Title	Due	Availability
D01	Annotation guidelines for sense tagging	m-6	Public, free
D02	Sense-annotation-tool	m-6	Public, free
D03	Subcorpus-1, sense tags and offsets or text	m-18	Public, current corpus license
D04	Subcorpus-2, sense tags and offsets or text	m-24	Public, current corpus license
D05	Subcorpus-3, sense tags and offsets or text	m-36	Public, current corpus license
D06	Knowledge rich word-sense disambiguation tool (WSD-1)	m-30	Public, free
D07	Supervised word-sense disambiguation tool (WSD-2)	m-30	Public, free
D08	Unsupervised word-sense disambiguation tool (WSD-3)	m-30	Public, free
D09	Final report	m-36	Public, free

For collaboration, we will set up email lists, a project Trac system with subversion and Wiki and we will have quarterly project meetings. All data and tools are shared within the project to stimulate discussion and communication. On the project website, we will start a forum to get early feedback from the research community and to organize public web-based contributions. Each member of the consortium is represented in the project management board. The consortium members are responsible for their own tasks and their own team. The overall coordination and reporting is done by the main contractor VUA, specifically by Prof. Vossen.

The project as a whole represents a major effort, but the risks for the project are low, while benefits are high. The products from both the manual and automatic methods of tagging will be useful for further research. Through double tagging and validated system-supported tagging, the quality of tagging can be continuously measured, and levels of quality can be guaranteed. Continuous logging, analysis, and monitoring by a post-doc, as well as regular editorial sessions provide further guarantees for the quality of the tagging. The largest risk for the project is the quantity of words that can be tagged. We have therefore derived a realistic estimate for the tagging time based on the pilot results and experiences in projects for other languages, to make sure that the student assistants can complete the job (see Appendix). These estimates are in accordance with the speed reported for the development of the English SemCor: 200 words per hour (Christiance Fellbaum, personal communication).

7.3 References (max 1 page)

- Agirre, Eneko & Oier Lopez de Lacalle. 2004. Publicly available topic signatures for all WordNet nominal senses. *Proceedings of the 4rd International Conference on Language Resources and Evaluations (LREC)*. Lisbon, Portugal.
- Agirre E., Aldezabal I., Etxeberria J., Iruskietia Quintian M., Izagirre E., Mendizabal K., Pociello E. 2006, A methodology for the joint development of the Basque WordNet and Semcor. *Proceedings of the 5th International Conference on Language Resources and Evaluations (LREC)*. ISBN 2-9517408-2-4. Genoa (Italy).
- Agirre E., and P. Edmonds (eds), 2006, Word Sense Disambiguation Algorithms and Applications Series: [Text, Speech and Language Technology](#) , Vol. 33 , 2006, ISBN: 978-1-4020-4808-1
- Atserias J., Villarejo L., Rigau G., Agirre E., Carroll J., Magnini B., Vossen P. [The MEANING Multilingual Central Repository](#). In: *Proceedings of the Second International Global WordNet Conference (GWC'04)*. ISBN 80-210-3302-9. Brno, Czech Republic. Enero, 2004.
- Fellbaum, Christiane, Martha Palmer, Hoa Trang Dang, Lauren Delfs, Susanne Wolf, 2001, [Manual and Automatic Semantic Annotation with WordNet](#), In: *WordNet and Other Lexical Resources*, NAACL 2001 Workshop, Pittsburg.
- Killgariff, Adam, 1999, 95% Replicability for manual word sense tagging, In: *Proceedings of the ninth conference on European chapter of the Association for Computational Linguistics*, Bergen, Norway. Pages: 277 - 278 ,1999
- Koeling, Rob, Diana McCarthy & John Carroll. 2005. Domain-specific sense distributions and predominant sense acquisition. *Proceedings of the Human Language Technology Conference and Conference on Empirical Methods in Natural Language Processing (HLT/EMNLP)*, 419–426.
- Magnini, Bernardo, Carlo Strapparava, Giovanni Pezzulo & Alfio Gliozzo. 2002. The role of domain information in word sense disambiguation. *Natural Language Engineering*, 8(4):359–373.
- Schrooten, W., and Vermeer, A. (1994). Woorden in het basisonderwijs: 15.000 woorden aangeboden aan leerlingen. *Studies in meertaligheid* Nr. 6. Tilburg: Tilburg University Press.

8. The quality and competence of the research team.

The consortium represents a combination of three areas of expertise:

- Computational lexicology
- Automatic word sense disambiguation
- Corpus annotation technology

UvT and UvA have developed state-of-the-art technology in word sense disambiguation. Their systems are supervised and/or need a training corpus. UvT has a long-standing expertise in using machine learning methods for automatic tagging and semi-automatic annotation of corpora. VUA is an internationally renowned player in the area of wordnets, ontologies, and computational lexical resources. VUA also developed unsupervised word-sense-disambiguation technology that uses knowledge from wordnet. All partners have a long-standing experience with language and text technology. VUA and UvA recently worked together in the Stevin project Cornetto, which yielded the Cornetto database, and all 3 partners collaborated in the NWO Middelgroot *startsubsidie* DutchSemCor.

Recent publications by the applicants (5 for each participating department/university):

VUA

1. Vossen P., Maks I., Segers R., Van der Vliet H. (2008). [Integrating lexical units, synsets and ontology in the Cornetto Database](#), in: *Proceedings of LREC 2008*, Marrakech, Morocco, May 28-30 May 2008.
2. Agirre E, Lopez de Lacalle O., Magnini B., Otegi A., Rigau G., Vossen P. (2008). [SemEval-2007 Task 01: Evaluating WSD on Cross-Language Information Retrieval](#), in: *CLEF 2007 (Cross Language Evaluation Forum)*. Also to be published as a Lecture Notes in Computer Science ([LNCS](#)), Vol. (fc.), Springer-Verlag 2008.
3. Vossen, P. (2008). [Linguistic knowledge for more precision, richer answers and flexible systems](#), in *Special Issue of Revue Francaise de Linguistique Appliquee (RFLA) on "Extraction d'Information: l'apport de la linguistique"*, Vol XIII 2008-1, June 2008. Revue Française de Linguistique appliquée, Paris.

4. Agirre E, Lopez de Lacalle O., Magnini B., Otegi A., Rigau G., Vossen P. (2007). [SemEval-2007 Task 01: Evaluating WSD on Cross-Language Information Retrieval](#), in: *Proceedings of SemEval 2007*, co-located with ACL-2007, June 23-24, 2007, Prague.
5. Buitelaar, P., Magnini, B., Strapparava, P. Vossen (2006). [Domain-Specific Word Sense Disambiguation](#). In: Eneko Agirre and Philip Edmonds (eds) [Word Sense Disambiguation - Algorithms and Applications](#). Springer, June.

UvA

1. Balog, K. and de Rijke, M. (2008). Non-Local Evidence for Expert Finding. In *ACM 17th Conference on Information and Knowledge Management (CIKM 2008)*, October.
2. Jijkoun, V. and de Rijke, M. (2008). Using Centrality to Rank Web Snippets. In: C. Peters, et al., editors, *Advances in Multilingual and Multimodal Information Retrieval*, LNCS 5152, September.
3. V. Jijkoun, V. Mahboob, K. Marx, M. and de Rijke, M. (2008). Named Entity Normalization in User Generated Content. In: *Proceedings of the second workshop on Analytics for noisy unstructured text data (AND 2008)*, pages 23-30, ACM.
4. Fissaha Adafre, S., Jijkoun, V. and de Rijke, M. (2007). Fact Discovery in Wikipedia. In: *2007 IEEE/WIC/ACM International Conference on Web Intelligence*, November.
5. Ahn, D., van Rantwijk, J. and de Rijke, M. (2007). A Cascaded Machine Learning Approach to Interpreting Temporal Expressions. In: *Proceedings NAACL-HLT 2007*, April.

UvT

1. Hoste, V., Hendrickx, I., Daelemans, W., and Van den Bosch, A. (2002). Parameter optimization for machine learning of word sense disambiguation. *Natural Language Engineering*, **8:4**, pp. 311-325.
2. Veenstra, J., Van den Bosch, A., Buchholz, S. and Daelemans, W. (2000). Memory-based word sense disambiguation. *Computers and the Humanities*, **34:1-2**.
3. Decadt, B., Hoste, V., Daelemans, W., and Van den Bosch, A. (2004). GAMBL: Generic algorithm optimization of memory-based WSD. In R. Mihalcea and P. Edmonds (eds.), *Proceedings of the Third International Workshop on the Evaluation of Systems for the Semantic Analysis of Text (Senseval-3)*, Barcelona, Spain, July 2004, pages 108-112.
4. Daelemans, W., and Van den Bosch, A. (2005). *Memory-based language processing*. Cambridge, UK: Cambridge University Press.
5. Hendrickx, I., Van den Bosch, A., Hoste, V., and Daelemans, W. (2002). Dutch word sense disambiguation: Optimizing the localness of context. In *Proceedings of the Workshop on word sense disambiguation: Recent successes and future directions (Senseval-2)*, Philadelphia, PA, USA, pp. 61-65.

9. Social relevance

The Dutch language is core to the Dutch information society. Technology development requires that information systems can be rooted in the Dutch language community and not just in English. Especially for people with lesser skills in English and technology, it is essential that research and technology development are ported from English to Dutch. A DutchSemCor corpus is key to this process. Intelligent language processing will increasingly become part of our daily life. Administrations and commercial systems will become more dependent on systems where citizens directly communicate with data servers. This communication needs to be in Dutch to allow equal and fair access to these services and to help lesser-skilled and elderly people. The STEVIN program of the Taalunie has resulted in a large text corpus and a semantic database. However, the two efforts have not been bridged: the text corpus has not been annotated with the semantic database, and the STEVIN program has now ended. Through this project we can complement the investment of STEVIN by producing a semantically annotated corpus, which will be the empirical basis for many new research and technology projects on the Dutch language.

10. National interest and access

SemCor and other semantically tagged corpora have been available for English for many years now. These corpora have been used for building supervised WSD systems, which up to now are the best performing systems, and on top of WSD systems many other applications can be built and research questions can be answered.

A semantically tagged corpus of Dutch can be used for training language models in technology systems differently from what has been done so far, just using large structurally annotated text corpora. It will provide valuable information on the interpretation of language in different contexts, which is a requirement

for the development of the language-technology of the future. This does not only relate to the usual text mining and question-answer systems, but also for automatic speech-recognition, machine translation, summarization, language-generation and language education. Such a corpus is essential for developing precise word-sense-disambiguation systems, data-mining software, and automatic concept and ontology acquisition tools for Dutch. Since the meanings are also related to English concepts in the Princeton wordnet, machine translation tools could be developed that use this corpus for aligning meanings and linguistics expressions with sense-tagged corpora in other languages. Studies on the usage of language in communication require the annotation of word meanings rather than just word forms. Especially more complex phenomena such as metaphoric and metonymic language use, irony, epistemic, and deontic expressions in relations to themes or cognitive modeling and linguistic framing in language can only be done if there is more information and knowledge about the basic semantic layers of interpretation. In a similar vein, contrastive studies across language systems can be done more properly when there is a common semantic anchor point to compare language structure. Finally, a proper discourse representation of language communication is also dependent on a more basic semantic layer of the compositional elements of the utterances. Obviously, such a corpus will generate empirical evidence for the complex relation between form and meaning that can be the basis for new theories and models, possibly reflected in more detailed and complex lexical resources that are fully corpus-based and compress an abstract representation or generalization of empirical evidence.

We expect new research and technology using this corpus in the following areas:

A. Fundamental research and enabling technology development:

1. (Cognitive-)Linguistic research on the relation between linguistic form and meaning: many claims on the relation between form and meaning have never been validated empirically because there has not been a large semantically-annotated text corpus for Dutch;
2. Linguistic research that relates sense frequencies to types of text and genres. This does not only include regular corpus-based research, but also contrastive studies where similar semantic content is related to different structural and genre properties. This can also be applied across different languages through the connection that is established with the English wordnet in the Cornetto database, and via English to any other language that is connected to the English wordnet;
3. Research on metaphoric and metonymic usage of language: as reported above we can only study the creative use of language if there is data on the basic and straight usage of language. This does not only provide a means to validate cognitive-linguistic models for metaphoric and metonymic usage but also for developing automatic ways to detect such usage and derive a possible interpretation for the deviant language usage;
4. Extremely complicated phenomena, such as modality, irony, sarcasm or even humor, play a role in most daily text. Such phenomena can only be studied empirically given solid statistical data on semantic usage of language in text;
5. Research on sentiment and attitude analysis of text: this type of research can reach new grounds if a more precise semantic interpretation of text is available. The Cornetto database and any corpus annotated with senses from this database can be used to derive more precise data on sentiments and opinions expressed in text;
6. Summarization and discourse representation technologies based on semantic coherence, co-reference relations; coherence is important for detecting which sentences present information on the same topics. Such information can be learned from a semantically annotated corpus;
7. Language generation technology in general: the corpus will enable study on variation in expressing the same or similar content in different words but also on what is the most frequently used way of expressing. This is extremely useful for language generation purposes;
8. Machine translation: the Global Wordnet initiative connects wordnets in many languages through a common index. This also applies to the Cornetto database. Likewise, we can align sense-tagged corpora across languages through a common semantic index as well and apply completely new ways of linguistic alignment to text. Current techniques are based on limited parallel corpora and structural word-by-word or constituent-by-constituent alignment. Through DutchSemCor, we can study other more loose ways of expressing content across languages.
9. Improving syntactic parsers: certain dependency constraints, e.g., PP-attachments, can only be derived by associating syntactic structures with semantic and conceptual patterns;
10. Improving speech recognition systems: current systems could rule out possible errors by building up semantic expectations that can be derived from the annotated corpus;

11. Named entity recognition software: locations (among which all geographical entities), time-expressions, organizations and individuals are usually detected on the basis of structural properties and so-called gazetteers. With DutchSemCor, these programs can also include semantic patterns for their detection, e.g., motion verbs with particular prepositions increase the likelihood that it involves a location or a path.
 12. Automatic lexicon and ontology acquisition: new words and concepts can be related to known words and concepts on the basis of the semantic, distributional and structural properties of the known words. A semantically tagged corpus directly relates these patterns to the meaning of the known words, thus enabling more precise and more effective learning of new words, new meanings of known words and their related concepts;
 13. Textual entailment is seen as a new technology to handle linguistic variation in expressing information. The corpus will allow us to directly derive this data in a robust and massive way;
 14. Data mining and fact extractions: these emerging technologies now use structural properties that can be detected relatively easy. Using the semantic patterns of DutchSemCor, we can derive more precise and new conceptual constraints for this technology. For example, we can learn text with words that refer to professions of people as a syntactic subject, what actions are carried out and can therefore be related to each profession and the people that have these roles.
 15. Machine-learning-based language technology: machine learning in general is hampered by sparse data and genre and domain dependency on the training data. A semantically tagged corpus allows us to study these phenomena and find new solutions. E.g., structural training data can be generalized to all words belonging to the same semantic class or the effects of structural variation can be compensated.
- B. End-user applications:
16. Language learning and language teaching: the semantic organization of the Cornetto database provides excellent opportunities for vocabulary learning, i.e., through known words and concepts, related concepts can be accessed that are not known or familiar. The corpus now also provides a rich database on how these words are used in this specific meaning;
 17. Tools for disabled or lesser-skilled people to use language (e.g., blind, deaf and dyslectic); In general these people have difficulty processing structural or perceptual properties of language, whereas they do not experience problems in conceptual modeling. Semantic models based on DutchSemCor can help to provide lesser choices in recognizing words or selecting words through semantic predictions;
 18. Information retrieval and question & answering systems: these systems suffer from recall and precision in various practical contexts where questions and available information cannot be matched directly. The corpus will yield new information for matching queries and questions to answers based on semantic interpretation;
 19. Semantic web technology building on automatic acquisition of meta data for large volumes of unstructured data, which is necessary for the development of practical systems in real situations where unstructured textual data dominate;
 20. Text classification systems are traditionally statistic systems that assign fixed labels to new text by comparing these with training examples. The training is based on relations between words and not on word meanings: the DutchSemCor corpus however makes it possible to discover the direct semantic structure that underlies these statistic word clusters. This makes it possible to derive more structured relations between text documents that are related to the same topics. This heavily depends on the semantics of events and change related to a dynamic reality. News, for example, reflects our view on the changing of the world within time-frames. These changes have specific temporal and causal relations. Such deep semantic analysis is helped through vast quantities of semantic data that will come available through this project.

To create a quick survey of the research interest in a DutchSemCor corpus, we sent out an e-mail with a support request to the participants of the workshop and other known research groups in the Netherlands. This mailing was sent out a week before the deadline of the proposal. The responders to the email are listed in the Appendix of the proposal.

11. Description of environment and implementation of the requested equipment

The three partners will collaborate through client-servers architectures. A dedicated server is needed to host the Cornetto database and another large server is needed to store the corpus. The VUA will provide the Cornetto database server as part of the matching. The UvA will provide a server for storing the corpora

and hosting the server for the sense-annotation tool. UvA and UvT need separate servers for training the WSD systems. The servers will be hosted by each partner and where necessary service can be accessed through http-protocols. Collaboration mainly depends on data processing and data exchange. No further special infra-structure is required.

12. Financial/technical aspects

12a Specification of the requested budget

Item description	Paid by NWO	Matching by applicants	Total
personel costs VUA	259,200	88,800	348,000
personel costs UvT	137,250	45,750	183,000
personel costs UvA	153,750	51,250	205,000
computing equipment VUA	0	3,000	3,000
computing equipment UvT	3,000	3,000	6,000
computing equipment UvA	3,000	3,000	6,000
Total	556,200	194,800	751,000

12b Expected breakdown in 3 years

Total costs in k€ (incl. matching)	year 2009	year 2010	year 2011
751,000	260,333	245,333	245,333

12c Do you expect financial assistance from another source? No

12d Personnel costs

As explained in the description of the work, most of the costs are involved with manual tagging and preparation of the text corpora. The resulting data are a valuable resource for research and technology development. A semantically tagged corpus for Dutch does not exist and can only be created by native speakers with sufficient training in the field. In the next table, we give a more precise motivation of the personnel and equipment costs.

Table 2: Budget overview

Budget	Nr .	Costs/ Year	Total fte p-p.	Total Costs	Prop. Subsidy	Subsidy	Matching	Proportion of matching
VUA								
student assistant	4	36,000	1.00	144,000	100%	144,000	0	
professor	1	120,000	0.60	72,000	50%	36,000	36,000	
wetenschappelijke programmeur	1	66,000	0.50	33,000	0%	0	33,000	
post-doc	1	66,000	1.50	99,000	80%	79,200	19,800	
servers	1	3,000		3,000	0%	0	3,000	
Subtotal				351,000		259,200	91,800	26.15%
UvT								
post-doc	1	66,000	1.67	110,000	75%	82,500	27,500	
wetenschappelijke programmeur	1	66,000	0.83	55,000	75%	41,250	13,750	
professor	1	120,000	0.15	18,000	75%	13,500	4,500	
servers	2	3,000		6,000	50%	3,000	3,000	
Subtotal				189,000		140,250	48,750	25.79%
UvA								
post-doc	1	66,000	1.67	110,000	75%	82,500	27,500	
wetenschappelijke programmeur	1	66,000	1.17	77,000	75%	57,750	19,250	

professor	1	120,000	0.15	18,000	75%	13,500	4,500	
servers	2	3,000		6,000	50%	3,000	3,000	
Subtotal				211,000		156,750	54,250	25.71%
Total				751,000		556,200	194,800	

Work plan and time schedule

Time-schedule	Participant	month 1-6	month 7-12	month 13-18	month 19-24	month 24-30	month 30-36
WP1 Corpus selection and preparation							
1.1 Collection of corpus and data processing	UvT	2					
1.2 Define corpus annotation format for storing weighed sense-tags	UvT	1					
1.3 Apply domain classifier to paragraphs	UvT	2					
1.4 Build (KWIC) indexes of word lemmas, incl. weighted sense tags and domain tags	UvA	2					
WP2 Sense annotation tool							
2.1 Functional and graphical design	UvA	1					
2.2 Access to the KWIC index and sorting options	UvA	1					
2.3 Access to Cornetto database, various rankings and votings, Wikipedia	UvA	1					
2.4 Storing of selected tags in corpus	UvA	1					
2.5 Logging of user actions	UvA	1					
2.6 Reporting tools (Interannotator agreement, tagging statistics)	UvA	1					
2.7 Web client with user verification	UvA	1					
WP3. Automatic tagging of subcorpus-1							
3.1 WSD-1: Build unsupervised WSD based on Cornetto database and other repositories	VUA	6					
3.2 WSD-1: run on subcorpus-1	VUA		2				
WP4. Manual verification and correction of the subcorpus-1 (250K words, double annotated)							
4.1 Annotation guidelines	VUA	2					
4.2 Correct and complete the tags in subcorpus-1	VUA		10	10			
4.3 Adapt Cornetto sense distinctions based on subcorpus-1	VUA		2	2			
WP5. Automatic tagging of subcorpus-2							
5.1 Adapt WSD-2 to use Cornetto and DutchSemCor	UvA	4	2				
5.2 Adapt WSD-3 to use Cornetto and DutchSemCor	UvT	4	2				
5.3 Build topic-signature extraction for tool with WSD-2	UvA		2	4			
5.4 Build topic-signature extraction for tool with WSD-3	UvT		2	4			
5.5 WSD-1: Run on subcorpus-2	VUA			2			
5.6 WSD-2: Train with subcorpus-1 and topic signature and run on subcorpus-2	UvA			2			

5.7 WSD-3: Train with subcorpus-1 and topic signatures and run on subcorpus-2	UvT			2			
WP6. Manual verification and correction of the subcorpus-2 (500K words, single annotation)							
6.1 Correct and complete the tags in subcorpus-2	VUA				10	10	
6.2 Adapt Cornetto sense distinctions based on subcorpus-2	VUA				2	2	
WP7. Automatic tagging of subcorpus-3							
7.1 Adapt WSD-2 using feedback	UvA				3		
7.2 Adapt WSD-3 using feedback	UvT				3		
7.3 Adapt topic-signature extraction for tool with WSD-2	UvA				3		
7.4 Adapt topic-signature extraction for tool with WSD-3	UvT				3		
7.2 WSD-1: Run on subcorpus-3	VUA					2	
7.3 WSD-2: Train with subcorpus-1&2 and run on subcorpus-3	UvA					4	
7.4 WSD-3: Train with subcorpus-1&2 and run on subcorpus-3	UvT					4	
WP8. Manual verification and correction of the subcorpus-3 (250K words, single annotation)							
8.1 Correct and complete the tags in subcorpus-3	VUA						8
WP9. Documentation and delivery							
9.1. Documentation of corpus and WSD-1	VUA						2
9.2 Documentation of WSD-2	UvA				1		
9.3 Documentation of WSD-3	UvT				1		
Total	136	30	22	26	26	22	10

(Different colors indicate different types of staff involved. Yellow: student assistants; green: scientific programmer; peach: postdoc. See Section 12d.)

Appendix: Results of the startsubsidie Workshop and Pilot

During the period July-August 2008 we have carried out a pilot study to tag a number of words (see Table 3). For this purpose, VUA developed a pilot annotation tool that accesses the Cornetto database to retrieve information on the meanings of a word and generates a list of all occurrences of a word in a fragment of the Corpus Gesproken Nederlands. All interactions during tagging are logged. Furthermore, a custom program was built to analyse the resulting log files. In the pilot, EU parliament discussions were indexed. Figure 1 below shows a screen dump of the tagging environment, with a search for the word “beeld.” The Cornetto meanings are given in the right-top window; the corpus occurrences in the bottom window. The left and right context of each occurrence is displayed in cells and the occurrences can be sorted on left context (retrograde), right context (alphabetic), the order in the corpus, and the sense tags. Annotators can select one or more senses in the Cornetto window, then select rows to which the sense applies, which will then be tagged with the selected senses.



The actual tagging was carried out in two rounds. First, we experimented with a number of random words to test the system and discuss the approach. Next, we made a selection of fourteen words across different parts of speech that were subsequently tagged by three trained people. The logs of these tagging efforts were processed to obtain average estimates of time and the degree of agreements for the tokens. Table 3 lists all the tagged words, where the agreed words are highlighted. For each word, the table shows the number of senses, the average time to tag all tokens, and the number of annotated instances. The annotations are differentiated per number of annotators who tagged the word occurrences (one, two, or three). The agreement column indicates the average agreement across the annotators.

Table 3: Effort and inter-annotator agreement for pilot study

Word	Nr. of senses	Avg. token duration in secs.	1 annotation	2 annotations	agreement	3 annotations	agreement
aardig	3	6	29				
beeld	5	10	176	261	39.08	180	30.28
beschaving	2	4	85	24	87.50		
bitter	5	7	60				
boek	5	9	207				
boos	2	5	21	15	33.33		
breken	9	11	50	79	42.41		
buigen	5	6	19	93	97.85	151	93.38
denken	6	1	1496	2510	70.78	651	68.69
dom	5	21	32	0			
dood	5	6	99	14	64.29		
druk	8	8	193				
enorm	2	2	744	248	80.24		
gaan	17	6	612	0			
gek	5	6	51				
geval	4	5	866				
grijpen	2	15	191	24	87.50		
groot	5	1	1767				
helder	7	3	285				
heten	3	10	16	81	35.80		
huis	5	9	4	82	31.71		
jong	4	1	130	382	59.42	447	69.80
klein	6	2	1774	272	36.76		
kop	8	5	4	25	56.00	66	43.10
kunnen	5	14	6				
man	3	2	975				
menen	4	11	160				
mild	3	5	0	2	0.00	47	53.90
neerleggen	8	6	60	53	47.17	76	39.47
oud	4	5	587	325	64.41		
positie	8	13	312	347	40.92	60	21.67
ruw	5	6	1	17	88.24	30	82.78
slag	18	2	339	73	86.30	54	51.85
sluiting	3	5	119	20	30.00	116	31.03
stom	3	13	6				
vies	4	42	0	6	75.00	0	
vuur	5	8	9	77	35.06		
weg	6	6	1121	129	39.53		
wrak	4	13	6	18	100.00		
zacht	7	5	114				
zeggen	4	14	210				
zwaar	4	6	201				

Agreement for a word (as a type) is calculated as follows:

- each assigned sense is scored by the number of annotations that were assigned to a token divided by the total number of annotators that handled the word: $AG(s) = s/a$, where s is the number of assignments of a sense and a is the number of annotators that tagged the word.
- the score for each token (word occurrence) is averaged by the scores of the assigned senses: $AG(token) = AG(s)/n_s$, where n_s is the number of assigned senses.
- the score for each type (word regardless of occurrence) is the average score for all tokens: $AG(type) = AG(token)/n_t$, where n_t is the number of tokens.

The average agreement for tokens with two annotators for all words is 57.17%. For 3 annotators this is 53.27. The average agreement for the words from the second round is 61.09 for two annotators and 52.62 for 3 annotators.³ Agreement quoted in the literature is around 80% for experts and about 65% for students. Agreement of 80% and higher is reached for a number of words but others have a much lower agreement. Either high or low agreement is reached and only rarely average agreement, which points to problems with certain words. Analysis of the disagreement showed that these low scores are mostly related to different strategies for difficult cases and errors in the Cornetto database. For example, in the case of “beeld,” one annotator selected sense 5 for the mental image based on the ontological label *Proposition* and the domain *psychology*, whereas the other annotator selected the 3rd meaning for that purpose and treated the 5th as concrete due to the synonym “plaatje.” To get consensus on how to handle such cases, it is necessary to go through a training phase. We expect that average agreement of 80% is feasible, but that it requires editorial sessions and discussions. These findings were also reported during the DutchSemCor meeting that was held in May-2008 in Amsterdam as a preparation to this proposal.

The average time spent on tagging during the pre-tagging phase is about 10 seconds, and in the second phase 7 seconds per word occurrence. This high speed is mostly due to the effective selection of examples by sorting on contextual clues. The tagging was carried out by highly trained experts. We expect that more time is needed for tagging by student assistants. Furthermore, we have not included the time needed to revise the Cornetto entries nor editorial meetings and sessions. The recorded time thus can be seen as the upper bound of tagging speed. We expect therefore that the real tagging speed to be considerably lower.

Another important aspect of the tagging is that annotators need time to understand the different meanings of highly-polysemous words. The more information is displayed, the more time is needed. This will increase the average time spent on tagging, but will also increase the quality of the tagging. During the DutchSemCor workshop it also became clear from experiences in other sense annotation projects that first cycles of work require editorial team discussion to achieve higher agreement and fix problems. Again, this will increase the time spent on tagging.

The workshop and the pilot also revealed that it is best to consider a word with all its meanings only once. Targeted tagging (all occurrences of a words are considered through a KWIC index) is therefore preferable to sequential tagging (every word in a text is tagged in the order of the text and the same word occurs at different places and needs to be considered many times). This makes it difficult to support the tagging with machine-learning tools that learn while people tag. If the project is set up in cycles of tagging, training and tagging, the annotators need to look at the same word many more times. We therefore concluded that a better project cycle would be to split the tagging process in a first phase of full manual tagging, and a second phase consisting of validating the automatic tagging. In the first phase, the annotators need to consider all the meanings of the words, and need to find sufficient example for each meaning. In the second phase, the annotators only consider the cases where the Word-Sense-Disambiguation systems have a low score or different systems have conflicting proposals. Those cases need to be validated or corrected, a decision task which is likely to take less time than the cognitively more demanding task of selecting a sense from scratch. Another difference is that for full manual annotation, we need to apply double tagging by two annotators so that we can derive inter-annotator agreement, which is exploited by the machine learning systems.

Evaluation of the pilot exercise and the disagreements further showed:

³ If we look at the total set of annotations (regardless of the words) then the agreement is higher: 65.05 and 59.59. This is due to certain words with high scores that contribute many tokens in total compared to others.

1. In certain cases the information on the word meanings and the contexts was not sufficient to make a choice, which can lead to errors. This needs to be improved in the actual annotation environment;
2. Special measures need to be taken to detect multiword phrases such as separable phrasal verbs (e.g., “uitbreken,” “aanbreken”) and idioms (e.g., “een lans breken”). A search for the general verb “breken” is now polluted with many of these cases. Possibly, occurrences of these should be flagged to exclude them from tagging, unless the multiword phrase is searched for;
3. The corpus is likely to be too restrictive to find sufficient examples for all senses of the selected words. Current experience is that too many examples are found for a few senses and none for others because of the genre and domain of the corpus;
4. We expect that selecting examples for senses using strong contextual filters will bias the machine learning systems. E.g. the query “positie van de vrouw” will give hundreds of examples for one meaning of “position,” but with a very restrictive and repetitive context. Therefore, we recommend the tagging of additional occurrences that are not well represented by such contexts. This will come at the costs of a higher tagging time;
5. The annotation interface should be able to constraint the search in many ways, e.g., select on part of speech, select on domain/genre, etc, to make sure that sufficient occurrences related to a specific meaning are found;
6. The interface should allow the user to tag word occurrences for figurative use, metonymic use (e.g., “hand” referring to the person that helps), specialization of meaning, a new meaning that is not listed, and label it with a question mark in case the annotator has no clue. The latter cases may lead to changing the Cornetto database, which should be done by the main editor that collects these labels and possible disagreements;
7. The rich experience of the pilot should be used to improve the design and ergonomics of the annotation tool user-interface.

We furthermore used the pilot estimates and experience to define a potential set of words to be covered in the DutchSemCor project and the amount of effort needed. The data for this are shown in the next two tables. As explained in the proposal, we want to focus on the hard cases. These are the most-frequent and most-polysemous words in Dutch. These are also the words that got special attention in the Cornetto database. All these words have been manually edited and checked.

Our selection will contain about 3,000 nouns, verbs and adjectives, which have 3.4 meanings on average, thus representing roughly 10,000 concepts in total. For comparison:

“The 121 most frequent English nouns have on average 7.8 meanings each and account for about 20% of word occurrences in real text (in the Princeton WordNet (Miller 1990), according to Ng and Lee (1996)).” (taken from Agirre and Edmonds 2006, page-1)

For a machine-learning system, 100 to 200 occurrences of each meaning need to be available as training data to attain disambiguation scores significantly above the baseline of choosing the most frequent sense, on average; the literature reports widely differing opinions. Hoste et al. (2002) remark that if a word expert is optimized by cross-validation with a known baseline, it does occur that as little as ten examples sometimes leads to a better machine-learned word expert than the baseline.

If we target 100 examples per word meaning/sense this would require a corpus of 1 million tokens to be manually tagged. We propose that this corpus of 1 million tokens to be divided into two parts, as shown in Table 5:

1. Initial full manual tagging (25%);
2. Validation tagging (75%);

In the case of full manual tagging, the annotators inspect all the meanings of a word and search for sufficient occurrences per meaning. The tagging needs to be carried out by two annotators per occurrence. At the start, we also need time for editorial changes to the database and reaching agreement on how to solve difficult cases. We estimate that it will take 18 seconds per token in this phase. To tag 250,000 tokens twice, the estimated effort is thus 20 person months, which are divided over 4 student assistants.

In the second phase validated tagging is applied to occurrences where the automated system score low or have different proposals. It is also applied to specific searches for word meanings that are rare in the basic corpus. The tagging will be faster since it does not require considering all the meanings of a word but slower since more efforts needs to done to get sufficient examples for rare meanings. We this estimate that this tagging also requires 18 seconds per token but they need to be tagged only once. We can thus cover 750,000 tokens in about 30 person months of effort, divided over 4 student assistants.

Table 4: Selection of types (words) and tokens (word occurrences)

Most frequent and most polysemous words in the Cornetto database	
2600	nouns
700	verbs
350	adj
3000	words in total
3.5	average nr of senses
10,000	estimated total nr of senses
100	tokens/sense required for machine learning based WSD
1,000,000	tokens needed in total
250,000	initial manual tokens 25%
750,000	validated manual tokens 75%

Table 5: Estimated effort based on the pilot of the startsubsidie DutchSemCor

	Nr. of tokens (word occurrences)	Nr. of annotators	Nr. of annotations in total	Sec/Token*	Person months (1500 hours/ year)
Initial tokens 25%	250,000	2	500,000	18	20.00
Validated tokens 75%	750,000	1	750,000	18	30.00
Size of the tagged corpus	1,000,000				50.00

* Based on pilot estimates, including editing the Cornetto database, This rating corresponds with the average tagging speed for the English SemCor: 200 words/hour (Christiane Fellbaum, personal communication).

Appendix: Program Workshop Dutch Semantic Corpus

Dutch SemCor Workshop

The Workshop DutchSemCor was held on Friday, May 16, 2008.

Workshop Topics and Aims

During this workshop we would like to discuss the development of a large, semantically tagged corpus for Dutch, comparable to the English SemCor and equivalents in other languages. Such a corpus is lacking and hampers Dutch language-technology development. Recently, a new large semantic database for Dutch has been built, the Cornetto database, which opens up the possibility to also deliver a semantically tagged corpus for Dutch. This corpus may play a key role in language technology research for Dutch, and also in linguistic and cognitive research that relates linguistic form to meaning. Combining the best of both worlds, the corpus will be tagged using a combination of automatic techniques and manual editing. Automatic tagging techniques include on the one hand supervised methods, which can be trained on already tagged subcorpora as training data, enabling them to tag other subcorpora, and on the other hand unsupervised techniques that rely on other sources such as the Cornetto database itself. It is to be expected that the manual editing of the corpus will feed back in the form of adaptations to the semantic database Cornetto. Possible characteristics of the envisaged DutchSemCor project are:

- a large corpus with sense-tags from the Cornetto database:
 - with a focus on the most-frequent, most-polysemous and domain-independent nouns, verbs and adjectives
 - excluding tagging of (fixed) collocational constructions The motivation for this restriction is that sense-selection for collocations requires firstly recognition of the collocation (IRME) and secondly representation of the meaning of the collocation in the Cornetto database. These seem to us matters for future research that will be enabled by the DutchSemCor corpus.
 - excluding metaphoric and metonymic uses The same holds for metaphoric and metonymic uses of words. Automatic detection and interpretation of these phenomena will be enabled by the DutchSemCor corpus as future research projects.
 - including some domain specialization as illustration Domain-based WSD is emerging and promising but is also relatively easy and thus avoids solving the hard domain-independent cases. We will include some specialized domain-tagging in the corpus but we expect that this issue can be solved without a DutchSemCor corpus, e.g. by automatically acquiring (unsupervised) domain corpora. These solutions are less language-dependent as well.
- a variety of sense-taggers based on different techniques: knowledge rich, supervised, unsupervised
- a sense-annotation tool that interacts with the Cornetto database

Programme

At the workshop, we present our preliminary plans and ideas about a project to build a Dutch SemCor. Furthermore, there are two invited presentations on the building of the English SemCor and a similar corpus for Basque. The afternoon session is used for discussion on issues related to the building and use of such a corpus. The following issues will be briefly introduced

- machine learning from semantically tagged corpora (prof. dr. Antal van den Bosch, ILK Research Group, Tilburg University)
- community tagging and wikipedia as additional resource (prof. dr. Maarten de Rijke, Intelligent Systems Lab, Universiteit van Amsterdam)

- language technology companies that may use such a corpus (dr. Leonoor van der Beek, Q-Go, Amsterdam)
- semantic tagging in D-COI and SONAR (dr. Paola Monachesi, Uil-OTS, Utrecht)
- linguistic research into the relation between form and meaning with the help of a semantically annotated corpus (Trijntje Pasma, Fac. der Letteren, VU, Amsterdam)

Agenda

10:00 - 10:30	Coffee
10:30 - 11:00	Project overview (Piek Vossen) ppt ⁴
11:00 - 11:30	Cornetto (Piek Vossen)
11:30 - 12:30	Basque SemCor (Eneko Agirre) ppt
12:30 - 13:30	Lunch
13:30 - 14:30	SemCor (Helen Langone) ppt
14:30 - 14:45	Coffee
14:45 - 16:15	Annotation tools for semantic tagging (Jakub Zavrel) <i>cancelled</i> Machine learning from semantically tagged corpora (Antal van den Bosch) ppt Community tagging and Wikipedia as additional resource (Maarten de Rijke)\ ppt Semantic Role Tagging in a Dutch Corpus (Paola Monachesi) ppt The use of a semantically tagged corpus by Language Technologies Companies (Leonoor van der Beek) ppt The use of a semantically tagged corpus for language and metaphor research (Trijntje Pasma) ppt
16:15 - 16:30	Discussion
16:30	Drinks

Organised by

Piek Vossen (Fac. der Letteren, Vrije Universiteit Amsterdam)

Isa Maks (Fac. der Letteren, Vrije Universiteit Amsterdam)

Antal van den Bosch (ILK Research Group, Tilburg University)

Maarten de Rijke (Intelligent Systems Lab, Universiteit van Amsterdam)

Invited Speakers and Talks

- **Helen Langone, Cognitive Science Laboratory, Princeton University** Helen Langone has a BS in Computer Science (1987) and MA in Linguistics (1998). She has been programming professionally for 20 years, the past 8 year of which have been in the lexicographic domain. She consults in the area of commercial lexicography, working on

⁴ The presentation slides from the meeting can be found at the website <http://www.let.vu.nl/onderzoek/projectsites/cornetto/dutchsemcor.html>.

dictionaries for Random House, OUP, CUP, and Lexico (dictionary.com). She has been a freelance consultant, and now part-time Technical Staff member, on the WordNet project on and off since 2002, contributing to a project to disambiguate the WordNet glosses.

SemCor and its { offspring, progeny, issue }: what can future projects learn from the experience of those who came before? Semantically annotated corpora marry dictionary definitions to words in context. Such a union brings together complementary aspects of the meaning of a word: syntagmatic and paradigmatic. A word is understood by the words it keeps company with in discourse, but also with respect to its relatives in its lexical "family tree". A semantically annotated corpus can therefore be a useful lexico-semantic asset for NLP researchers who are building and training systems for WSD, Q&A, Summarization, Information Retrieval and Information Extraction. Semcor, which the Brown corpus was semantically tagged to WordNet, is the grandfather of semantically annotated corpora. The disambiguated WordNet gloss project and the Basque Semcor are its latest progeny. Many issues that arose for Semcor will be encountered by its successors, and pitfalls that one project trips on will lie in wait for another. Targeted vs. sequential tagging, granularity of the senses, tagging to a static or moving target...how these questions answered can forestall certain problems but invite others. The question I will consider in this talk is: What can future projects learn from the experience of those who came before?

- **Eneko Agirre, University of the Basque Country, UPV/EHU, Basque Country** Eneko Agirre received his Ph.D. (1999) and B.A. (1990) in Computer Science by the University of the Basque Country, M.Sc. (1991) in Cognitive Science by the University of Edinburgh. He is currently a lecturer in the Computer Science Faculty of the University of the Basque Country since 1995. He has published over ninety refereed articles and conference papers in Natural Language Processing, mainly in the areas of Lexical Knowledge Acquisition, Word Sense Disambiguation and Semantic Processing. He has been the co-editor of the book "WSD: Algorithms and applications". He is a member of the program committees for major conferences like GWC, ACL, HLT, CONLL, TSD, RANLP and EMNLP, and has organized several workshops in major conferences (ACL and EACL). He is a usual reviewer for other major conferences including IJCAI, and several journals including LRE, NLE, JSL, ACM Computing Surveys, ACM TOIS and IEEE Intelligent Systems. He participates in the KNOW and KYOTO projects, and has participated in HERMES and MEANING. He has organized the SemEval-2007 semantic evaluations competition and workshop.

A methodology for the joint development of the Basque WordNet and Semcor This talk describes the methodology adopted to jointly develop the Basque WordNet and a hand annotated corpora (the Basque Semcor). This joint development allows for better motivated sense distinctions, and a tighter coupling between both resources. The methodology involves edition, tagging and refereeing tasks. The talk will review the guidelines for taggers and wordnet editors, and present the interfaces used.

Location

Hortus Botanicus Amsterdam
Plantage Middenlaan 2a,
Amsterdam, 020 - 6259021
www.dehortus.nl

Appendix: Adhesieverklaringen naar aanleiding van e-mail, verstuurd op 25 Augustus jongstleden.

Original e-mail replies can be supplied if requested.

25-08-2008

Geachte heer, mevrouw,

Ik wil graag mijn steun betuigen aan de onderzoeksaanvraag voor het project DutchSemCor dat als doel heeft om een Nederlands corpus van 1 miljoen woorden semantisch te taggen met betekenissen uit de lexicaal-semantische database Cornetto. Een dergelijk corpus kan een belangrijke rol spelen in het onderzoek en de (taal)technologieontwikkeling waar ik bij betrokken ben. Hiermee kunnen nieuwe onderzoeksrichtingen ontwikkeld worden. Het corpus en de afgeleide technologie zijn belangrijk om het Nederlands als communicatietaal voor de toekomst te behouden en om ervoor te zorgen dat mensen die het Engels onvoldoende beheersen gelijkwaardige toegang houden tot diensten.

Met vriendelijke groeten,

Naam: Iris Hendrickx

Organisatie: Universiteit van Antwerpen

Contact informatie: iris.hendrickx@ua.ac.be

Hoofdvactiviteit: Onderzoek

Beoogd onderzoek met DutchSemCor resultaten: Bouwen van WSD tool voor het Nederlands, en deze gebruiken als informatiebron voor mijn onderzoek naar coreferentie resolutie.

Naam: Emiel Krahmer

Organisatie: Universiteit van Tilburg

E-mail: e.j.krahmer@uvt.nl

Hoofdvactiviteit: Onderzoeker

Beoogd onderzoek met DutchSemCor resultaten: Onderzoek naar het detecteren en gebruiken van semantische overlap tussen zinnen (in het verlengde van het Stevin Daeso project, zie <http://daeso.uvt.nl/>)

Naam: Prof. Dr. Frieda Steur

Organisatie: Lessius/KULeuven

Contact informatie: frieda.steurs@lessius.eu

Hoofdvactiviteit: hoogleraar, departementshoofd

Beoogd onderzoek met DutchSemCor resultaten: taalcontrastief onderzoek (Nederlands vergelijken met andere talen), onderzoek op getagd materiaal is essentieel in de vertaaltechnologische toepassingen waar wij ons mee bezig houden.

Naam: Veronique Hoste

Organisatie: LT3 language and translation technology team, Hogeschool Gent, Associatie Universiteit Gent

Contact informatie: veronique.hoste@hogent.be

Hoofdvactiviteit: corpusgebaseerd taaltechnologisch onderzoek (<http://veto.hogent.be/lt3>)

Beoogd onderzoek met DutchSemCor resultaten: automatische desambiguering van woordbetekenissen (WSD) voor het Nederlands

Naam: Laura Hollink

Organisatie: Vrije Universiteit Amsterdam

Contact informatie: laurah@cs.vu.nl
Hoofdactiviteit: Universiteir Docent
Beoogd onderzoek met DutchSemCor resultaten: Het toegankelijk maken van de collecties van Nederlandse cultureel erfgoed instellingen, > zoals musea, bibliotheken en het Nederlands Instituut voor Beeld en Geluid, door middel van 'semantic web' applicaties.

Naam: Gosse Bouma
Organisatie: Informatiekunde, Rijksuniversiteit Groningen
Contact informatie: g.bouma@rug.nl
Hoofdactiviteit: corpustaalkunde, informatie-extractie
Beoogd onderzoek met DutchSemCor resultaten: ontwikkelen van algoritmes voor disambigueren van woordbetekenis, voor gebruik in information retrieval.

Naam: Gertjan van Noord
Organisatie: Alfa-informatica/Informatiekunde, Rijksuniversiteit Groningen

Naam: Aäron Jansen
Organisatie: Vrije Universiteit Amsterdam / Stagiair Q-go
Contact informatie: aaron.jansen@gmail.com
Hoofdactiviteit: Masteronderzoek (afstudeeronderzoek)
Beoogd onderzoek met DutchSemCor resultaten: Het verbeteren van het (semi-)automatisch vinden van synonieme termen uit grote tekstcorpora.

Naam: H. Coenders
Organisatie: Uitgeverij het Spectrum
Contact informatie: harry@spectrum.nl
Hoofdactiviteit: lexicografie

Naam: Prof. dr. Willy Martin
Organisatie: E. Prof. Lexicologie, VU Amsterdam
Contact informatie: w.martin@let.vu.nl
Hoofdactiviteit: projectleider ANNA (= Afrikaans-Nederlands, Nederlands-Afrikaans woordenboekproject) en voorzitter NL-Term (Vereniging voor Nederlandse Terminologie)
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Naam: Prof dr Gerard Steen
Organisatie: Afdeling taal en communicatie, onderzoeksgroep 'Language, cognition, communication'
Contact informatie: zie signature
Hoofdactiviteit: leider bovenstaande onderzoeksgroep, leider vici-programma 'Metaphor in discourse'
Beoogd onderzoek met DutchSemCor resultaten: onderzoek naar metaforiek in het nederlands, uitkomsten vergelijken met soortgelijk onderzoek uit vici-programma op basis van soortgelijk al bestaand corpus voor het engels

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Hoofdactiviteit: ontology mapping
Beoogd onderzoek met DutchSemCor resultaten: text mining
--
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Naam Wilbert Spooren
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Contact informatie: w.spooren@let.vu.nl
Hoofdactiviteit: hoogleraar Taal & Communicatie
Beoogd onderzoek met DutchSemCor resultaten: diverse corpusanalyses over subjectiviteit, tekststructuur en inhoudsanalyse in geschreven en gesproken tekst. Toelichting: veel van het huidige onderzoek is gebaseerd op handmatige corpusanalyses. Dat maakt de 'coverage' van de onderzoeksbevindingen gering en de analyses vatbaar voor codeerbetrouwbaarheidsproblemen. Grootschalige automatische analyse van semantische verschijnselen die gerelateerd zijn aan tekststructuur zullen een belangrijke oplossing vormen voor dit probleem.

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Hoofdactiviteit: onderzoek op het terrein van taal- en spraaktechnologie
Beoogd onderzoek met DutchSemCor resultaten: taalmodellering

Naam: Frank Van Eynde
Organisatie: Centrum voor Computerlinguïstiek Kath. Universiteit Leuven
Contact informatie: Blijde-Inkomststraat 13, Leuven
Hoofdactiviteit: Onderzoek op het gebied van de formele en computationele taalkunde met speciale aandacht voor de toepassing ervan in taal- en spraaktechnologie
Beoogd onderzoek met DutchSemCor resultaten: training van semantische tagger

Naam: Vincent Van Asch
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Hoofdactiviteit: Wetenschappelijk onderzoeker

Naam : Remco van Veenendaal
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Hoofdactiviteit : Projectleider TST-Centrale
Beoogd onderzoek met DutchSemCor resultaten:
De TST-Centrale zal het beheer en de distributie van de projectresultaten van DutchSemCor verzorgen. Als projectleider van de Centrale voor Taal- en Spraaktechnologie (TST-Centrale, www.inl.nl/tst-centrale) steun ik DutchSemCor van harte, omdat eindresultaten van dit project via de TST-Centrale voor hergebruik beschikbaar gesteld zullen worden.

Naam: Carole Tiberius
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Hoofdactiviteit: Onderzoek
Beoogd onderzoek met DutchSemCor resultaten: Automatic Text Analysis / Text Mining